

Software Project Management (IT414)

Course placement: Software Project Management is an elective course for the final year B.Tech, second year M.Tech, MScIT and MScDS programs.

Course format: 3 hours' lecture and 2-hour laboratory every week.

Course content:

Practical use of software project management requires understanding of core software design principles, related managerial aspects, an influx of new ideas, and the courage to improve traditional approaches for handling evolving scenario of IT industry. As a software engineer and an early career manager, you need to know the methods, workflows and tools to handle continuously growing complexity and shortened project management cycles. You must be able to work in teams to build high-quality software. In this course, we will introduce the basic concepts of software project management along with some contemporary industry practices. You will also get a hands-on on project documentation, feasibility analysis, risk handling techniques, etc. that are used when implementing complex software projects through different case studies.

Books and Literature:

- Walker Royce. Software Project Management: A Unified Framework.
- Kathy Schwalbe. Information Technology Project Management.
- Stellman & Greene. Applied Software Project Management.
- Richard Thayer, Edward Yourdon. Software Engineering Project Management.
- Bob Hughes and Mike Cotterell. Software Project Management.
- Steve McConnell. Code Complete: A Practical Handbook of Software Construction.
- M. Lewrick, P. Link, and L. Leifer. The Design Thinking Playbook.

Assessment Method: 6-8 lab assignments, a group project, class tests, and end-semester examination.

Course Outcomes:

Software Project Management theory introduces students to fundamental principles of software project management as well as risk management principles along with some related activities carried out in large scale IT software projects. It also covers key topics including estimation, project planning, scheduling, project reviews, process improvement, etc. activities involved in managing software projects. This course also address topics related to handling outsourced projects, project failures etc. This course also prepares students for higher level specialization in software project management courses.

After successful completion of the course the student will have the ability to,

- Understand, analyze, plan, manage, and implement software projects (P1, P2, P12).

- Apply project management principles for implementation / handling software projects using various tools and techniques (P3, P5, P11).
- Plan, monitor, track, and review software projects in a large group setup (P9, P11).
- Create software project management documents, detailed project plans, schedules, along with necessary software implementation strategy. (P10, P11)

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
√	√	√		√				√	√	√	√

Lecture Schedule

Topics / Description	No. of Lectures
Software Project Planning	4
Software Estimation	4
Software Project Scheduling, and Reviews	5
Software Requirements, Design, and Testing	3
Software Project Failures	3
Management and Leadership in Software Companies	4
Software Process Improvement	5
Software Risk Management	4
Introducing Design Thinking for Project Management	2
Management of Outsourced Software Projects	3
Industry Practices for Managing large-scale IT projects	3

Grading components and examination pedagogy

- Lab assessment – 50%
 - Weekly Lab Assignments (20%)
 - Case Studies (10%)
 - Group Project (20%)
 - Design, documentation, and implementation
 - Two Review Group Presentations – Mid-term and Final
- Class Tests (2) – 10%
- End-Semester Examination (40%)
 - Question types – theory, problem solving, case-study based, application, diagram type.